FARMERS' MARKET PRICE AND SALES TRACKER

Project Report

Team Members

1. M. Lohith — 2620040024
2. S. N. Darshan — 2620090099

1. Abstract

The Farmers' Market Price and Sales Tracker is a Java-based application concept designed to help users record crop details, update market prices, track quantities sold, and calculate sales amounts. It brings basic market and sales information into one simple program, making record keeping easier and reducing manual calculations. The project demonstrates how fundamental Java programming concepts can be applied to a practical agricultural-market problem.

2. Introduction

Farmers need to keep track of crop prices and sales to manage their produce effectively. When information is maintained manually, records can become difficult to organize and calculations may take extra time. This project proposes a straightforward menu-driven Java program that stores and displays crop and sales details during use. It is intended as a beginner-friendly example of using programming to solve a real-life record-keeping problem.

3. Problem Statement

Maintaining crop names, quantities, prices, and sales records manually can be inconvenient and may lead to calculation errors. Users also need a simple way to update prices and review saved records. The project addresses these needs through a single program with menu options for entering, updating, calculating, and viewing market information.

4. Objectives

- Record crop names, quantities, and prices.
- Allow market prices to be updated when they change.
- Record the quantity of produce sold.
- Calculate the total sale amount automatically.
- Display saved crop and sales records.
- Demonstrate basic Java programming through a practical example.

5. Main Features

Crop details: Enter the crop name, available quantity, and price per unit.

Price updates: Change the recorded market price when required.

Sales recording: Enter the quantity of crop sold.

Automatic calculation: Calculate the sale amount from quantity and price.

Record viewing: View the records entered during the program session.

6. Technologies and Java Concepts Used

Variables: Store crop names, prices, and quantities.

Scanner: Accept input from the user.

if-else and switch: Process menu choices and control program actions.

Loops: Repeat the menu until the user selects Exit.

Methods: Organize the program into smaller, reusable tasks.

ArrayList: Keep multiple records in a collection while the program runs.

7. Working Procedure

1. The program starts and displays a menu of available operations.
2. The user selects an option from the menu.
3. The Scanner class accepts the required input.
4. Java statements perform the selected operation or calculation.
5. The result is displayed to the user.
6. The menu appears again until the user chooses to exit.

8. Example Calculation

Suppose a user records a tomato sale of 20 kg at ■30 per kg.

Total Sale Amount = Quantity Sold × Price per kg

Total Sale Amount = 20 × ■30 = ■600

Therefore, the total value of the tomato sale is ■600.

9. Benefits

- Makes crop and sales record keeping easier.
- Reduces repeated manual calculations.
- Keeps entered information organized in one program.
- Helps users review crop prices and sales details.
- Provides practical experience with basic Java concepts.

10. Limitations and Future Enhancements

In its current beginner-level design, records stored in an ArrayList are available only while the program is running; permanent storage is not included. Future versions could save data in files or a database, provide price-history reports, support searches and stock alerts, and display trends through charts. These are possible improvements rather than features confirmed in the current project.

11. Conclusion

The Farmers' Market Price and Sales Tracker is a simple Java project that demonstrates how programming can help organize crop prices and sales information. Its menu-driven structure supports entering crop details, updating prices, recording sales, calculating totals, and viewing records. The project is suitable for beginners and can be expanded in the future with permanent storage and more advanced market-analysis features.

Prepared from the project presentation. Features such as permanent database storage and advanced analytics are listed as future enhancements, not existing capabilities.